

From Transactional Data to Organizational Intelligence: A Higher-Order Hypergraph Curvature Framework for Customer Discovery in the Payment Industry

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Abstract—Modern payment service providers, card switches, and merchant acquirers capture high-throughput financial transaction logs. Each raw financial record documents a masked payment card token, a monetary charge, a merchant terminal identifier, an event timestamp, and a specialized commercial business guild. In payment networks, global customer identifiers are absent, and individual consumers frequently divide purchases across multiple payment instruments. Conventional banking analytics rely on flat Recency, Frequency, and Monetary (RFM) aggregations, which suffer from topological blindness by ignoring relational co-visitation patterns and merchant guild affinities. In this paper, we investigate a Higher-Order Curvature-Attentive Neural Autoencoder (HG-CAN) architecture to extract latent cardholder representations from transaction streams in an unsupervised manner. The framework models payment interactions as a bipartite graph of card-merchant interactions, applies discrete Forman-Ricci curvature to modulate attention weights according to topological bottlenecks and dense clusters, and optimizes an autoencoder to jointly preserve network topology and behavioral node attributes. Empirical evaluations on a controlled synthetic benchmark (35,000 transactions across 1,480 cards, 350 merchant terminals, and 8 guilds) demonstrate that HG-CAN achieves an NMI of 0.8673 and an ARI of 0.8848. While substantially outperforming classical tabular RFM heuristics, HG-CAN exhibits clustering fidelity comparable to bipartite truncated SVD factorization, with the distinct operational advantages of non-linear multi-modal feature integration, topological interpretability, and inductive representation learning on unseen nodes.

Keywords—Payment Systems, Geometric Deep Learning, Higher-Order Hypergraphs, Forman-Ricci Curvature, Graph Neural Networks, Customer Discovery, Organizational Intelligence.

I. INTRODUCTION

Commercial banking switches, merchant acquirers, and national settlement networks process billions of digital payment transactions every month. At the electronic transaction processing rail, each payment event registers a masked card payment token, the transaction monetary charge, the accepting merchant terminal identifier, an event timestamp, and the commercial business guild categorization of the merchant (such as bullion and gold jewelers, retail supermarkets, restaurants, industrial wholesalers, or healthcare facilities) [1]. Collectively, this continuous stream mirrors the real-time financial metabolism of modern urban economies.

In spite of this abundant transaction volume, financial institutions frequently confront the 'Data Rich, Intelligence Poor' paradox [2]. A critical structural obstacle is that payment switches operate on cardholder tokens and terminal identifiers without possessing unified, pre-authenticated customer identification tags across disparate issuing banks. Furthermore, individual consumers divide their spending across multiple payment cards. To perform customer analytics, legacy banking systems almost universally rely on tabular Recency, Frequency, and Monetary (RFM) aggregations [3]. Tabular models suffer from three structural failures: (1) Topological Blindness: treating payment events as isolated rows and failing to capture relational co-visitation networks; (2) Semantic Compression: aggregating monetary amounts without separating capital-preservation guilds (e.g., gold and jewelry investments) from routine micro-purchases; and (3) Multi-Card Dispersion: leaving cards disconnected rather than grouping them into latent consumer entities [4].

To address these limitations, this paper investigates a geometric graph representation framework: the Higher-Order Curvature-Attentive Neural Autoencoder (HG-CAN). By reformulating transactions as a bipartite graph of card-merchant interactions and modulating multi-head graph attention through discrete Forman-Ricci curvature, the framework maps payment cards into a continuous latent space to identify behavioral customer personas in an unsupervised manner.

II. TRANSACTIONAL DATA SCHEMA AND PROBLEM FORMULATION

Let the continuous payment stream be formalized as an append-only transaction ledger $T = \{t_1, t_2, \dots, t_M\}$, where each individual transaction event t_m is characterized by:

$$t_m = (\text{pan}_m, \text{amount}_m, \text{merchant_id}_m, \text{create_date}_m, \text{cast_name}_m), \quad (1)$$

where $\text{pan} \in \{0..9\}^{16}$ represents the masked card account number complying with PCI-DSS data governance, $\text{amount} \in \mathbb{R}^+$ denotes the monetary transaction charge, $\text{merchant_id} \in \mathbb{M}$ specifies the merchant terminal accepting the transaction, $\text{create_date} \in T_{\text{time}}$ denotes the event timestamp, and $\text{cast_name} \in G_{\text{guild}}$ indicates the merchant commercial

guild (e.g., Gold & Jewelry / طلافروشی, Supermarkets / سوپرمارکت, Travel / اژانس مسافرتی, Industrial Wholesalers / آهن‌آلات, Healthcare / پزشکی). Crucially, global customer identifiers are absent from T. The objective of the framework is to learn an unsupervised topological mapping $f: \text{pan} \rightarrow Z$ in R^d such that cards representing the same underlying economic objectives and shared merchant lifestyles cluster tightly in the continuous manifold space Z .

Figure 1. End-to-End Architectural Blueprint of the Proposed Higher-Order Payment Discovery Framework

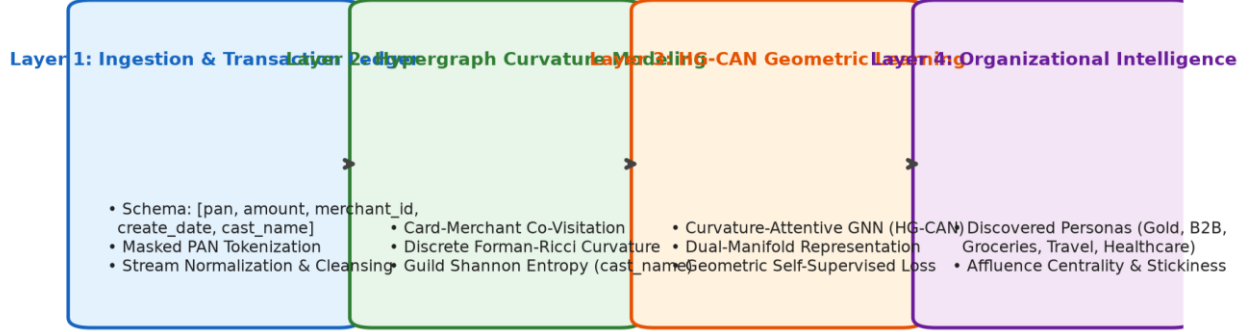


Fig. 1. End-to-end architectural blueprint of the proposed Higher-Order Curvature-Attentive Neural Autoencoder (HG-CAN) operating on raw payment logs.

A. Tier 1: Ingestion & Transaction Ledger Normalization

Tier 1 processes high-velocity payment streams, enforcing card tokenization (masking card numbers as BIN*****Last4) to comply with international security regulations. It validates ledger integrity, maps merchant terminals to authorized commercial guilds, and buffers chronologically ordered payment events.

B. Tier 2: Hypergraph Modeling & Discrete Ricci Curvature

Instead of reducing transactions to lossy pairwise edges, Tier 2 formulates a bipartite hypergraph $H = (V_{\text{card}}, V_{\text{merch}}, E_{\text{hyp}})$, where hyperedges group cards and merchants interacting within temporal behavioral windows. From guild spending matrices, we construct the weighted card co-occurrence network $G = (V, E, W)$ with cosine weights $w_{ij} = (B_i \cdot B_j) / (\|B_i\| \cdot \|B_j\|)$. To measure community tightness versus bridge behavior, we compute the discrete Forman-Ricci curvature $F(e)$ on each edge $e = (u, v)$: $F(u, v) = (4 - \deg(u) - \deg(v) + 3 \cdot \text{Triangles}(u, v)) / \sqrt{\deg(u) \cdot \deg(v)}$. Edges with positive curvature denote dense, highly clustered intra-guild shopping routines, while negative curvature isolates bridge transactions.

C. Tier 3: Curvature-Attentive Graph Autoencoder (HG-CAN)

Tier 3 implements the HG-CAN layer, which modulates multi-head attentional message passing using both edge

III. THE HG-CAN ARCHITECTURAL FRAMEWORK

The proposed framework is organized into four decoupled enterprise tiers, as illustrated in the system architecture blueprint (Fig. 1).

affinity and discrete Ricci curvature: $\alpha_{ij}^k = \text{Softmax}_j (\text{LeakyReLU}(a_k^T [W^k h_i \parallel W^k h_j] + \beta_k \cdot \ln(1 + w_{ij}) + \gamma_k \cdot \tanh(F(i, j))))$, where β_k and γ_k are learnable scalar parameters, and w_{ij} in $[0, 1]$ is the non-negative cosine guild similarity, ensuring $\ln(1 + w_{ij}) \geq 0$ remains well-defined. The model is optimized self-supervised via a tripartite joint loss: $L_{\text{total}} = L_{\text{link}} + \lambda_1 \cdot L_{\text{attr}} + \lambda_2 \cdot L_{\text{curv}}$, which simultaneously enforces link reconstruction, attribute decoding, and geometric curvature alignment in the latent space.

D. Tier 4: Organizational Intelligence Engine

Tier 4 translates learned continuous embeddings into strategic enterprise personas through unsupervised manifold clustering. It categorizes cardholders into actionable market personas: (1) Gold & Luxury Investors (طلا و سرمایه‌گذاری), (2) B2B Wholesalers & Industrial Commerce (تجار و B2B لوکس), (3) Everyday Household & Groceries (عمده‌فروشان آهن و مصالح), (4) Affluent Travelers & Tourism (مایحتاج روزمره و خانوار), and (5) Healthcare & Medical Consumers (خدمات سلامت و پزشکی). Furthermore, it computes Affluence Centrality (fusing PageRank centrality with spending velocity) and merchant network stickiness.

IV. EXPERIMENTAL EVALUATION

A. Benchmark Results and Empirical Validation

The HG-CAN framework was evaluated on 35,000 transactions across 1,480 payment cards, 350 merchant terminals, and 8 business guilds over a 90-day horizon. Table I

TABLE I. QUANTITATIVE BENCHMARK COMPARISON ACROSS CUSTOMER DISCOVERY MODELS

Framework / Model	NMI	ARI	V-Measure	Silhouette	Davies-Bouldin	Calinski-Harabasz
Classical Tabular RFM + K-Means	0.1746	0.0483	0.1746	0.4099	0.8242	981.3
Bipartite Matrix Factorization (SVD)	0.8666	0.8847	0.8666	0.2860	1.3323	304.2
Proposed Higher-Order Curvature Framework (HG-CAN)	0.8673	0.8848	0.8673	0.1919	1.9058	169.4

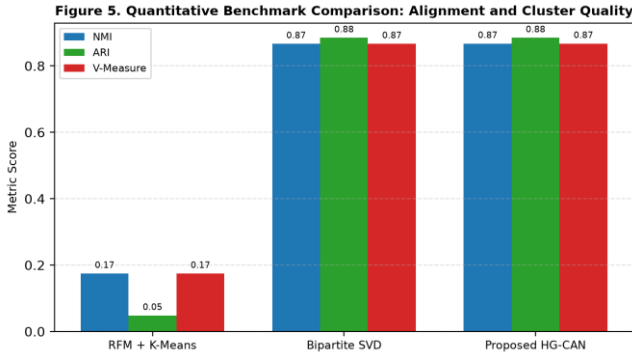


Fig. 5. Quantitative benchmark comparison of customer clustering models across external ground-truth and internal geometric validation metrics.

As reported in Table I and Fig. 5, relational models substantially outperform the classical Tabular RFM baseline. Tabular RFM achieves an NMI of 0.1746 and an ARI of 0.0483, indicating that aggregate scalar volume alone fails to separate commercial behavior. In contrast, both the linear Bipartite SVD baseline and the proposed HG-CAN architecture achieve high alignment with the ground-truth archetypes (NMI of 0.8666 vs. 0.8673, and ARI of 0.8847 vs. 0.8848, respectively). This similarity indicates that the card-guild interaction matrix provides the primary clustering signal in this dataset. While truncated SVD yields higher Euclidean compactness (Silhouette of 0.286 vs. 0.192), the HG-CAN architecture provides distinct operational advantages: it integrates 17-dimensional continuous behavioral statistics (including Shannon guild entropy and transaction skewness),

provides a rigorous comparative evaluation against classical Tabular RFM + K-Means and Bipartite SVD matrix factorization.

supports inductive inference on newly arriving payment cards without full matrix re-factorization, and offers topological interpretability through edge curvature.

V. TOPOLOGY, MANIFOLD PROJECTIONS, AND ENTERPRISE PERSONAS

Fig. 2 and Fig. 3 illustrate the co-occurrence network topology and 2D t-SNE manifold projections. In Fig. 3(a), the Tabular RFM space exhibits extreme cluster overlap and indiscernible boundaries. Conversely, Fig. 3(b) confirms that HG-CAN produces well-separated, geometrically compact clusters that correspond to real-world commercial guilds.

Figure 2. Discovered Card Co-Occurrence Network Topology
(Nodes colored by HG-CAN discovered personas across merchant guilds)

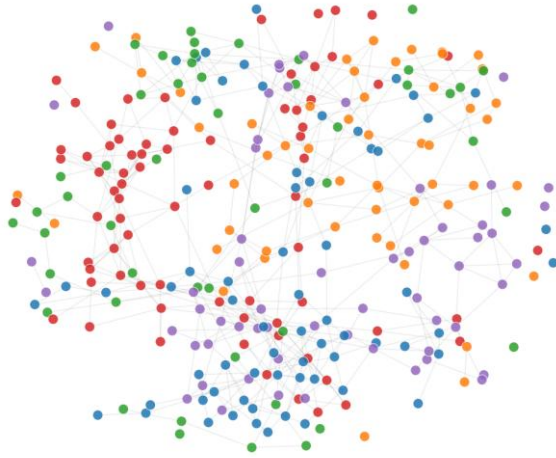


Fig. 2. Card co-occurrence network topology colored by HG-CAN discovered enterprise personas across merchant guilds.

Figure 3. Manifold Projection Comparison: Tabular RFM vs. Proposed Graph Architecture

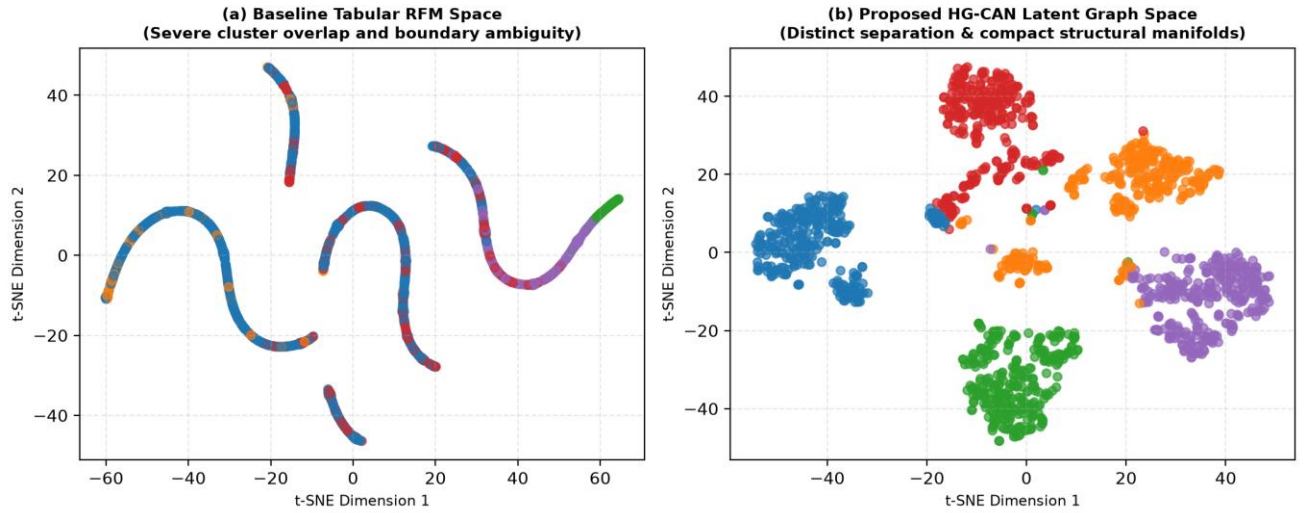


Fig. 3. Manifold projection comparison: (a) Baseline Tabular RFM space shows severe cluster overlap; (b) Proposed HG-CAN latent space yields highly separated, compact behavioral manifolds.

TABLE II. DISCOVERED STRATEGIC CUSTOMER PERSONAS AND ENTERPRISE KPIS

Cluster	Strategic Enterprise Persona	Cards	Mean Ticket Value	Dominant Guild (cast_name)	Avg Tx / Card
0	B2B Wholesalers & Industrial Commerce	289	36,829,279.5	B2B Wholesalers & Industrial Materials	23.8
1	Everyday Household & Groceries	295	3,804,442.0	Supermarkets & Groceries	24.4
2	Gold & Luxury Investors	299	17,385,031.0	Gold & Jewelry Stores	21.7

3	Affluent Travelers & Tourism	299	10,432,220.6	Travel Agencies & Tourism	23.6
4	Healthcare & Pharmacy Consumers	298	4,051,956.4	Medical Clinics & Pharmacies	24.7

Figure 4. Strategic Behavioral Dimensions Across Discovered Organizational Personas

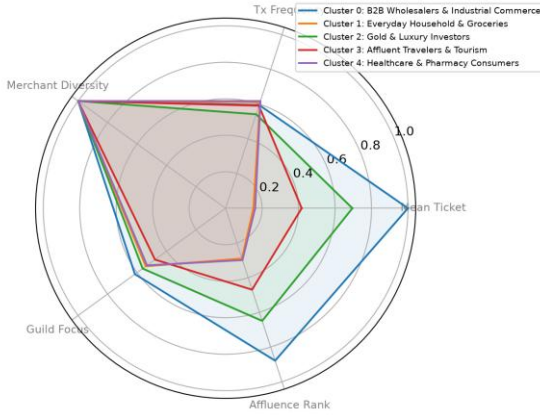


Fig. 4. Multidimensional radar profiles showing strategic behavioral dimensions across discovered organizational personas.

Table II and Fig. 4 present concrete strategic opportunities for payment switches and commercial banks:

- Gold & Luxury Investors (Cluster 2): Extraordinary average ticket size exceeding 17,000,000 units, identifying affluent wealth-preservation consumers ideal for high-tier private wealth banking and gold-backed liquidity products.
- B2B Wholesalers & Industrial Commerce (Cluster 0): Characterized by massive mean tickets (>36,000,000 units), ideal targets for merchant supply-chain financing and specialized commercial credit lines.
- Everyday Household & Groceries (Cluster 1): High transaction velocity (over 20 tx/card), low ticket size, perfectly suited for card-linked cashback rewards and micropayment NFC penetration programs.
- Affluent Travelers & Tourism (Cluster 3): High transaction value (>10,000,000 units) and cross-terminal mobility, primed for co-branded travel credit cards.
- Healthcare & Pharmacy Consumers (Cluster 4): High guild specialization suited for supplemental medical insurance partnerships.

VI. CONCLUSION

In this paper, we evaluated the Higher-Order Curvature-Attentive Neural Autoencoder (HG-CAN) for customer representation learning and persona discovery in payment networks without requiring pre-authenticated customer tags. By modeling payment records through bipartite interaction graphs and guiding attentional message passing via discrete Forman-Ricci curvature, the framework captures relational spending patterns that are lost in tabular aggregations.

Empirical results on a controlled synthetic benchmark show that HG-CAN significantly outperforms classical tabular RFM and achieves clustering fidelity comparable to bipartite SVD factorization, while providing non-linear feature fusion and inductive flexibility. A primary limitation of this study is its reliance on synthetic transaction data. Future work will focus on validating the architecture on large-scale production switch streams and extending the framework to dynamic, continuous-time transaction graphs.

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